

ARTEM MAKLAKOV

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EDUCATION

Moscow Institute of Physics and Technology (MIPT, PhysTech)

2023 - Present

B.S. in Applied Mathematics and Computer Science, expected June 2027

- Cumulative GPA: 4.84/5.00

Relevant courses:

- Advanced Machine Learning, Efficient Deep Learning, Computer Vision, Optimization Methods, Matrix Analysis, Probability Theory, Mathematical Statistics

School No. 57, Moscow

2021 - 2023

Graduated with a gold medal. Advanced physics class.

RESEARCH PROJECTS

Adaptive Scaling for GLUON in Neural Network Training

Sep 2026 - Present

BRAIn Lab, Moscow Independent Research Institute of Artificial Intelligence

- Investigate adaptive scaling mechanisms for GLUON inspired by Adam and AdEMAMix.
- Study how the scaling matrix defines the underlying norm and how adaptive scaling affects convergence analysis.

Market Microstructure and Continuous-Time Exchange

January 2026 - May 2026

Industry project with Sber

- Investigated a multidimensional extension of market microstructure invariance to correlated assets, focusing on matrix-valued cross-impact and liquidity relations.
- Implemented order-clearing and liquidity-integration components for a continuous-time exchange prototype inspired by Kyle and Lee's "Toward a Fully Continuous Exchange."

INDUSTRY EXPERIENCE

Huawei Technologies - Research & Development

Software Engineer Intern, DSP Compilers team

July 2025 - Present

- Develop optimization methods for production LLVM-based DSP compilers, focusing on instruction scheduling and register allocation.
- Developed and evaluated ML-guided compiler heuristics using profiling and performance measurements.
- Apply combinatorial and constraint optimization to compiler optimization problems using Unison and MiniZinc.

Huawei Open Days, AI for Code team

August 2024

- Researched the impact of Retrieval-Augmented Generation on code generation quality.

OPEN SOURCE

LLVM Upstream Contribution

- Merged contribution to LLVM upstream, worked with LLVM maintainers during the review process

HONORS AND AWARDS

All-Russian Olympiad of schoolchildren (Physics)

April 2022

The final stage prize-winner

International Experimental Physics Olympiad (IEPHO)

November 2019

Two times bronze medal

November 2020

Level I National Olympiads

2020-2023

Multiple winner in physics and mathematics

TEACHING AND LEADERSHIP

System Programming and Compiler Development course

- Teaching Assistant for two years; Senior Mentor in the third year.
- Mentor students in systems programming and compiler development, review assignments, and provide technical guidance.

Academic Outreach

- Represented MIPT's School of Applied Mathematics and Informatics at the final stage of the All-Russian Olympiad of Schoolchildren in Physics.
- Organized and proctored the Level I "Phystech" Physics and Mathematics Olympiad in Yekaterinburg.

MIPT Freshman Mentoring

- Mentor first-year students and organize orientation and group activities.

SELECTED SOFTWARE PROJECTS

"57Lang" programming language with x86 compiler

December 2023 - May 2024

Tools: C/C++, ASM

- Designed and implemented a programming language with a complete frontend, IR, optimization, and x86-64 backend.
- Generated native ELF executables directly, achieving ~50× faster execution than the previous emulator-based implementation.

TECHNICAL SKILLS

- | | | | | |
|----------|-----------------|-----------|------------|--------------------------|
| • C/C++ | • Deep Learning | • PyTorch | • MiniZinc | • Functional Programming |
| • Python | • Optimization | • Rust | • CUDA | • LLVM/MLIR |

LANGUAGES

- **Russian:** Native
- **English:** C1 - IELTS Academic 8.0/9, October 2025
- **Japanese:** Beginner, currently studying towards JLPT N4
- **German:** Beginner, currently studying at MIPT

INTERESTS AND HOBBIES

- **Ice Hockey:** Play in an amateur team in the Night Hockey League.